Project Report

datacamp

Recipe Site Traffic Optimisation

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BUSINESS INTELLIGENCE

ACCOMPLISHED 5 MARCH 2026

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1.Introduction

Statistics plays a crucial role in understanding uncertainty and real world phenomena. This project applies core concepts of descriptive statistics to a recipe site dataset to study what makes a recipe popular.
the predictive model must correctly identify popular recipes at least 80% of the time.
Main goals of the project are :

- Clean and validate the dataset
- Explore patterns with descriptive analyses and visualizations
- Build predictive models that generalize to new data
- Test statistical hypotheses about recipe features and popularity

2.Dataset Description

Dataset : Recipe_site_traffic_2212
Source : DataCamp

2)1.Dataset Overview

- The dataset contains information about different recipes, including their ingredients, nutritional values, preparation details, and whether they were popular or not.
- It is used to analyze patterns and relationships between recipe characteristics and their popularity.
- The data is in a CSV file, small in size (~40 kB).

3.Data Validation

	recipe	calories	carbohydrate	sugar	protein	category	servings	high_traffic
0	1	NaN	NaN	NaN	NaN	Pork	6	High
1	2	35.48	38.56	0.66	0.92	Potato	4	High
2	3	914.28	42.68	3.09	2.88	Breakfast	1	NaN
3	4	97.03	30.56	38.63	0.02	Beverages	4	High
4	5	27.05	1.85	0.80	0.53	Beverages	4	NaN

By examining the first rows of the dataset, we can observe that some recipes have missing values for key nutritional features such as calories and sugar, which need to be handled during the data cleaning process.

```
Missing values (initial):
recipe            0
calories         52
carbohydrate     52
sugar            52
protein          52
category          0
servings          0
```

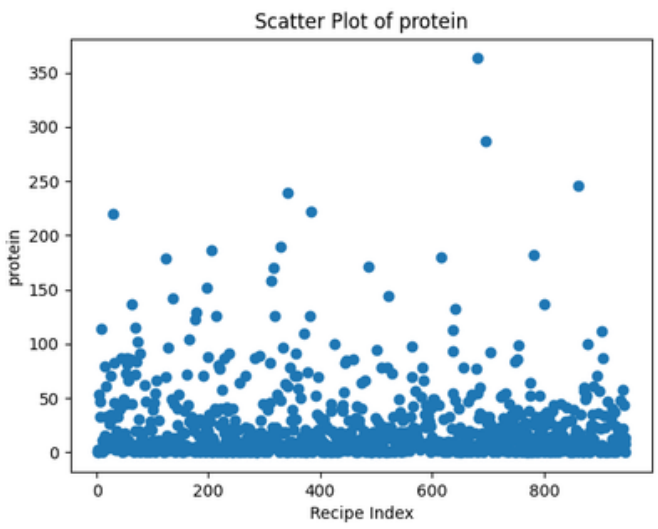
Dropp rows where all ['calories', 'carbohydrate', 'sugar', 'protein'] were missing.

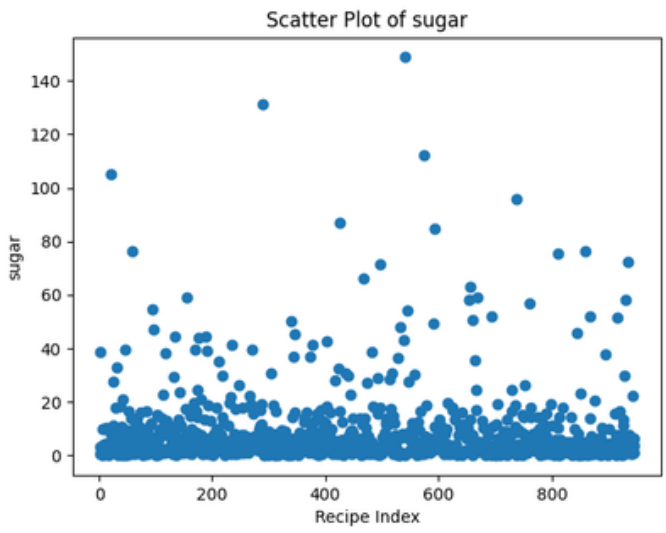
```
Target distribution after mapping:
high_traffic
1    574
0    373
```

In the high_traffic column, we treated NA values as low traffic represented by 0, and the actual high_traffic represented by 1, as I didn't find any clear mention of low_traffic in the data.

```
Summary statistics:
            recipe     calories  carbohydrate       sugar     protein   category servings high_traffic
count   947.000000   895.000000    895.000000  895.000000  895.000000        947      947         574
unique         NaN          NaN           NaN         NaN         NaN         11        6           1
top            NaN          NaN           NaN         NaN         NaN  Breakfast        4        High
freq           NaN          NaN           NaN         NaN         NaN        106      389         574
mean    474.000000   435.939196     35.069676    9.046547   24.149296        NaN      NaN         NaN
std     273.519652   453.020997     43.949032   14.679176   36.369739        NaN      NaN         NaN
min       1.000000     0.140000      0.030000    0.010000    0.000000        NaN      NaN         NaN
25%     237.500000   110.430000      8.375000    1.690000    3.195000        NaN      NaN         NaN
50%     474.000000   288.550000     21.480000    4.550000   10.800000        NaN      NaN         NaN
75%     710.500000   597.650000     44.965000    9.800000   30.200000        NaN      NaN         NaN
max     947.000000  3633.160000    530.420000  148.750000  363.360000        NaN      NaN         NaN
```

Looking at the percentile distribution, we can see that some variables, such as sugar and calories, have very high maximum values compared to their upper percentiles, indicating the presence of outliers, by investigating mean vs median we can clearly see that these columns are right skewed, which need further investigation by plotting these columns distribution.

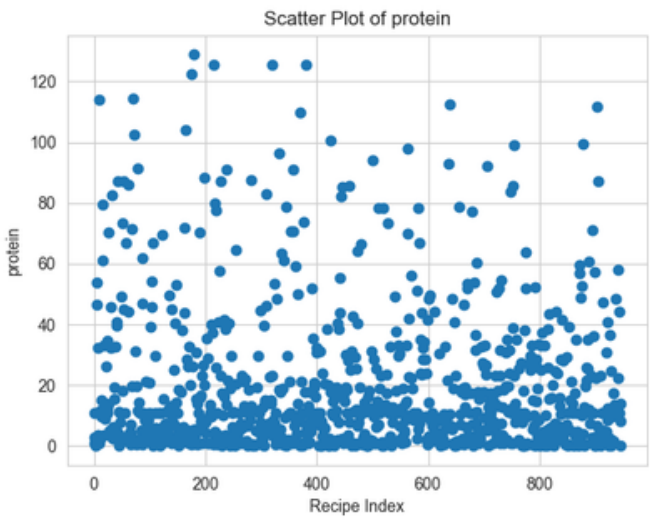


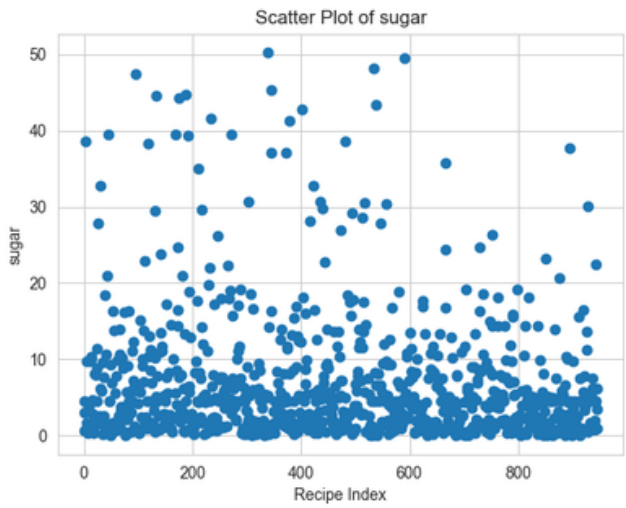


From these two scatter plots, we can clearly see that some points are far above the main cluster of values.

```
Filtered 15 rows from 'calories' above 3 std dev (1752.87)
Filtered 17 rows from 'carbohydrate' above 3 std dev (163.88)
Filtered 23 rows from 'sugar' above 3 std dev (50.55)
Filtered 19 rows from 'protein' above 3 std dev (130.35)
Fitting 5 folds for each of 4 candidates, totalling 20 fits
```

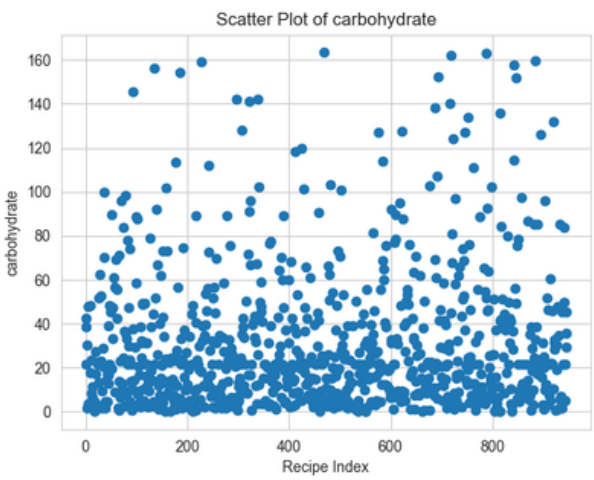
I flagged out rows where the value is more than 3 standard deviations above the mean.



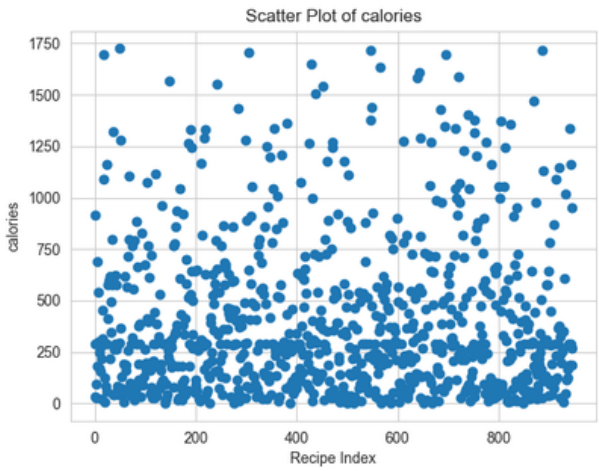


4.Exploratory Analysis

The initial four scatter plots, showed some outlier values, but after sorting them out, the plot show a more correlated values.

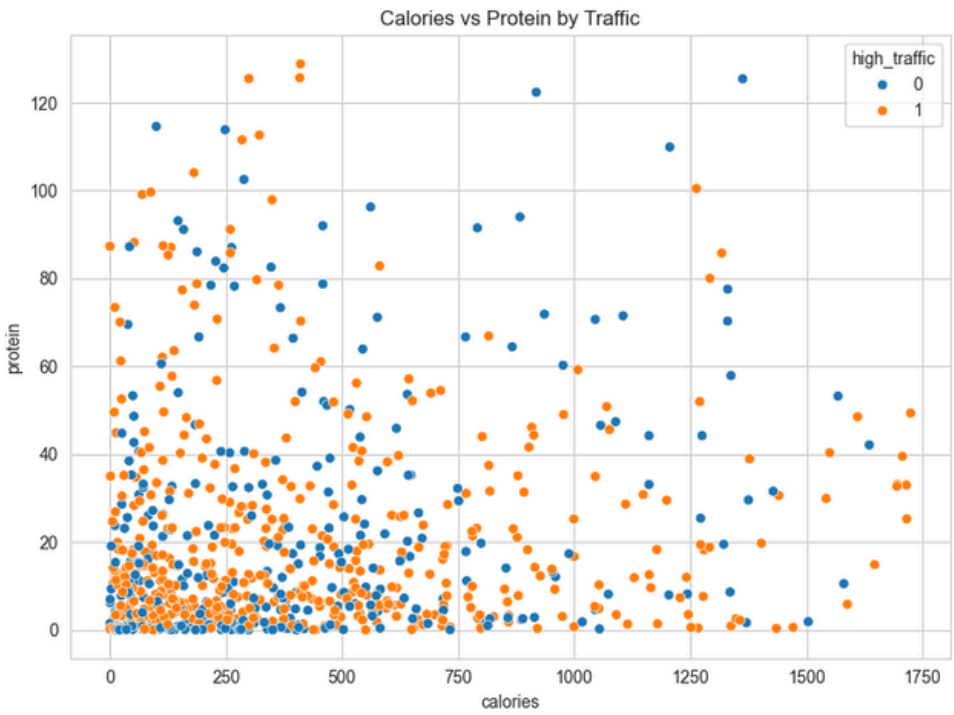


The carbohydrate plot, show that most of the values fall between 0 and 100 g.



The carbohydrate plot, show that most of the values fall between 0 and 1000 g.

=> there is some minor indications of different scaled values, which might bias our model.



- No strong relationship between calories and protein.
- Most of the recipes fall into a small amount of calories and protein.

5.Model Development

- Our model will solve a classification problem, of whether a recipe can be considered as high or low traffic.
- This is a binary classification problem because we are predicting one of two possible outcomes.
- The team specifically wants the model to correctly identify high-traffic recipes at least 80% of the time. This means high recall for the positive class (high_traffic = 1) is a priority better to show a recipe that is likely popular than to miss it.
- I will drop the recipe identifier as the model doesn't need it.
- we will scale all the values (mean = 0, std = 1) before fitting them into the model, as our exploratory analysis showed that our features are on different scale.
- I used a preprocessing pipeline, to ensures consistency and reproducibility.
- I divided the dataset into training (80%) and test set (20%), with keeping the same proportion of high and low traffic recipes in both train and test sets with balanced weights.

5.1 Logistic Regression and Gradient Boosting Model:

- I fitted a Logistic Regression model, with different hyperparameter combinations.
- The scoring metric focus on recall with Gridsearch CV .
- The CV method use shuffling and stratifying for better results.
- At the end the model shows the hyperparameters that gave the highest recall

6.Model Evaluation

```
Logistic Regression (default 0.5) Results
              precision    recall  f1-score   support

           0       0.65      0.76      0.70        70
           1       0.82      0.73      0.77       105

    accuracy                           0.74       175
   macro avg       0.74      0.75      0.74       175
weighted avg       0.75      0.74      0.75       175

Accuracy: 0.7428571428571429
```

```
Gradient Boosting (default 0.5) Results
              precision    recall  f1-score   support

           0       0.65      0.79      0.71        70
           1       0.83      0.71      0.77       105

    accuracy                           0.74       175
   macro avg       0.74      0.75      0.74       175
weighted avg       0.76      0.74      0.75       175

Accuracy: 0.7428571428571429
```

6.1 Logistic Regression Results

- Model is slightly better at precision for high traffic than recall. You catch most predicted popular recipes correctly, but some high-traffic recipes are missed.

6.2 Gradient Boosting Results

- Likewise, Gradient Boosting slightly favors precision over recall. It predicts popular recipes, but misses slightly more actual high traffic recipes.

=> close to the 80% target but not quite there. ≈

- **Logistic Regression: slightly better recall, fewer popular recipes missed**
- **Gradient Boosting: slightly better precision, predictions more confident**

6.3 threshold optimization to reach recall (≥ 0.80)

- We will Test 99 thresholds (0.01 → 0.99) and compute for each : Recall, Specificity, Precision, Accuracy and Select the threshold that achieves recall ⩾ 0.80 and maximizes specificity among those thresholds.

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```
Gradient Boosting best threshold: 0.410000000
Gradient Boosting metrics at that threshold:
Recall (sensitivity): 0.819047619047619
Specificity (true negative rate): 0.614285714
Precision: 0.7610619469026548
Accuracy: 0.7371428571428571
Confusion matrix:
 [[43 27]
 [19 86]]
```

```
Logistic Regression best threshold: 0.46
Logistic metrics at that threshold:
Recall (sensitivity): 0.8
Specificity (true negative rate): 0.7
Precision: 0.8
Accuracy: 0.76
Confusion matrix:
 [[49 21]
 [21 84]]
```

- Gradient Boosting slightly improves recall
- But Logistic Regression:
  has better precision
  has better specificity
  has better accuracy

=> The better model here is Logistic Regression (threshold = 0.46)

- **It meets the recall requirement (>=0.80) while producing fewer false positives.**

7.The Business Metric Analysis

7.1 Metric definition

Recommended metric:

- Recall for high traffic recipes
- The homepage manager wants to ensure that most popular recipes appear on the homepage, therefore missing a popular recipe is worse than showing a few less popular ones.

So the business objective becomes:

Recall >= 80%

7.2 How the business should monitor it ?

The business should monitor recall by comparing:

- Recipes predicted as high traffic by the model
- Recipes that actually generate high traffic

Steps :

1. Run the model to select recipes for the homepage.
2. Collect actual traffic data after publication.
3. Label recipes as high traffic vs low traffic.
4. Compute recall.

Retrain model if recall drops below target.

8.Summary and Recommendations

- The objective was to build a model to identify recipes that generate high traffic. Two classification models were evaluated: Logistic Regression and Gradient Boosting.
- Both models achieved similar performance threshold 0.5. However, the recall for high traffic recipes was around 73%, which did not meet the business objective of identifying at least 80% of high traffic recipes.
- To address this, threshold tuning allowed both models to achieve the target recall. Logistic Regression reached a recall of 0.80 at a threshold of 0.46, while Gradient Boosting achieved 0.82 at a threshold of 0.41.
- Despite the slightly higher recall of Gradient Boosting, Logistic Regression produced higher precision (0.80), higher specificity (0.70), and higher overall accuracy (0.76) compared to Gradient Boosting. This indicates that Logistic Regression identifies popular recipes reliably while generating fewer incorrect recommendations.
- Based on the evaluation results, Logistic Regression with a threshold of 0.46 is recommended as the final model.
- To further improve performance, the model should be periodically retrained as new traffic data becomes available.